10.26 Dirac Spinors in Practice

§10.25 classifies the spinor representations; LecNote16 pp. 11–17, LecNote17 pp. 1–10 and LecNote18 pp. 3–12 compute with them. A page number without a LecNote refers to the last LecNote named before it. The computations raise their own questions:

- "$\psi^a\chi_a = -\psi_a\chi^a$, yet $\chi\psi = \psi\chi$ (LecNote16 pp. 12, 15). When do I get a sign, and which matrix acts on ψ^a? Where is the $\sigma^\mu\bar\sigma^\nu + \sigma^\nu\bar\sigma^\mu = -2g^{\mu\nu}$ that LecNote17 p. 2 takes 'from last lecture'?"
- "Why $\bar\Psi = \Psi^\dagger\beta$ and not $\Psi^\dagger$? β 'numerically equals γ^0 but meaning is different' (LecNote17 p. 5). Different how? And what are C and a Majorana field for (pp. 6–8)?"
- "What does $u_s(\mathbf{p})$ look like after the boost? Why $\bar u u = 2m$, why does LecNote18 p. 8 project with $\bar u\gamma^0$ and $v(-\mathbf{p})$, and what is the Gordon identity of p. 11? Is helicity the same as chirality (pp. 7–8)?"

In short: signs come from the antisymmetry of ε and from Fermi statistics, and an upper index carries the inverse transpose. β is the invariant pairing of $\Psi^\dagger$ with Ψ (the representation is not unitary); γ^0 is the time component of a vector. C maps u to v. Plane waves follow from four rest-frame spinors and one boost; helicity is chirality only at $m = 0$.

Conventions. Srednicki's and the lecture notes': $g = \mathrm{diag}(-,+,+,+)$; $\sigma^\mu_{a\dot a} = (I, \boldsymbol\sigma)$, $\bar\sigma^{\mu\dot a a} = (I, -\boldsymbol\sigma)$; $\varepsilon^{12} = \varepsilon_{21} = +1$ for the undotted and the dotted spinor metric (as a matrix, $\varepsilon^{ab} = i\sigma_2$, the ε of §10.25); $\gamma^\mu = \left(\begin{smallmatrix} 0 & \sigma^\mu \\ \bar\sigma^\mu & 0 \end{smallmatrix}\right)$, $\{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}$, $\beta = \left(\begin{smallmatrix} 0 & I \\ I & 0 \end{smallmatrix}\right)$, $\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \mathrm{diag}(-I, I)$, $S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$, $\not p = \gamma^\mu p_\mu$; the Levi-Civita symbol has $\epsilon^{0123} = +1$, $\epsilon_{0123} = -1$; the Dirac equation is $(-i\not\partial + m)\Psi = 0$. With $U(1+\delta\omega) = 1 + \frac{i}{2}\delta\omega_{\mu\nu}M^{\mu\nu}$, $[\Psi, M^{\mu\nu}] = \mathcal{L}^{\mu\nu}\Psi + S^{\mu\nu}\Psi$ (LecNote16 p. 4). Boosts: we use Srednicki's $K_i = M^{i0}$ (LecNote16 p. 9), so $K^i = S^{i0}$ on spinors, and $e^{i\eta\hat{\mathbf{n}}\cdot\mathbf{K}}$ ($\omega_{i0} = -\omega_{0i} = \eta n_i$) takes $(m, \mathbf{0})$ to $m(\cosh\eta, \hat{\mathbf{n}}\sinh\eta)$. §10.3 and §10.25 define $K_i = M^{0i}$, the negative: their $e^{i\eta\hat{\mathbf{n}}\cdot\mathbf{K}}$ is our $e^{-i\eta\hat{\mathbf{n}}\cdot\mathbf{K}}$.

Spinor indices: the rules

Indices are raised and lowered from the left, $\psi^a = \varepsilon^{ab}\psi_b$, $\psi_a = \varepsilon_{ab}\psi^b$ ($\varepsilon_{ab}\varepsilon^{bc} = \delta_a{}^c$), dotted ones alike; undotted indices are contracted ${}^a{}_a$, dotted ones ${}_{\dot a}{}^{\dot a}$: $\chi\psi \equiv \chi^a\psi_a$, $\chi^\dagger\psi^\dagger \equiv \chi^\dagger_{\dot a}\psi^{\dagger\dot a}$ (LecNote16 pp. 12–15). Minus signs have two independent sources. *Flipping a contraction*, $\chi^a\psi_a = \varepsilon^{ab}\chi_b\psi_a = -\chi_b\varepsilon^{ba}\psi_a = -\chi_b\psi^b$, reorders nothing and holds for numbers too (p. 12). *Fermi statistics*, $\psi_a(x)\chi_b(y) = -\chi_b(y)\psi_a(x)$ (p. 15; the reason is in §10.27), is the other. Together they give $\chi\psi = -\psi_a\chi^a = \psi\chi$ and

$(\chi\psi)^\dagger = \psi^\dagger\chi^\dagger = \chi^\dagger\psi^\dagger$; two flips and one exchange give

$$\psi^\dagger\bar\sigma^\mu\chi = \psi^\dagger_{\dot a}\varepsilon^{\dot a\dot b}\varepsilon^{ab}\sigma^\mu_{b\dot b}\chi_a = \psi^{\dagger\dot b}\sigma^\mu_{b\dot b}\chi^b = -\chi\sigma^\mu\psi^\dagger \tag{10.26.1}$$

for anticommuting fields (for commuting spinors the sign is +).

Which matrix acts on which index. If $\psi_a \to L_a{}^b\psi_b$, the invariance of $\psi^a\chi_a$ forces $\psi^a \to [(L^T)^{-1}]^a{}_b\psi^b$; indeed $\varepsilon^{ab}L_b{}^c\varepsilon_{cd} = [(L^T)^{-1}]^a{}_d$ since $\det L = 1$. Dotted indices work the same way (the table below). So the generator on an upper index is *minus the transpose*. In index language a transpose is "moving both indices": for traceless M, $\varepsilon^{ac}\varepsilon_{bd}M_c{}^d = M_b{}^a$, while the unit matrix gives $\varepsilon^{ac}\varepsilon_{bd}\delta_c{}^d = -\delta^a{}_b$. That is the "−" which, says LecNote17 p. 10, index placement "takes care of".

field	matrix	generator	name in the notes
ψ_a	L	$S_L^{\mu\nu}$	L (LecNote16 p. 7)
ψ^a	$(L^T)^{-1} = \varepsilon L\varepsilon^{-1}$	$-(S_L^{\mu\nu})^T$	
$\psi^\dagger_{\dot a}$	L^*	$S_R^{\mu\nu} = -(S_L^{\mu\nu})^*$	R (LecNote16 p. 9, LecNote17 p. 9)
$\psi^{\dagger\dot a}$	$(L^\dagger)^{-1} = \varepsilon L^*\varepsilon^{-1}$	$-(S_R^{\mu\nu})^T = (S_L^{\mu\nu})^\dagger$	R (§10.25)

THE MATRIX ACTING ON EACH INDEX POSITION, $L \in SL(2,\mathbb{C})$

σ AND ε IDENTITIES

Srednicki derives these identities in §§35–36, in the notation used here (LecNote16 pp. 13–16): $\bar\sigma^{\mu\dot a a} \equiv \varepsilon^{ab}\varepsilon^{\dot a\dot b}\sigma^\mu_{b\dot b} = (I, -\boldsymbol{\sigma})$, σ^μ with both indices raised ((35.19)–(35.20)); the invariance of σ^μ ((35.9), the $X \to LXL^\dagger$ of §10.25) and of $\bar\sigma^\mu$, which makes $\psi^\dagger\bar\sigma^\mu\chi$ a vector; $\sigma^\mu_{a\dot a}\sigma_{\mu b\dot b} = -2\varepsilon_{ab}\varepsilon_{\dot a\dot b}$ and $\operatorname{tr}\bar\sigma^\mu\sigma^\nu = -2g^{\mu\nu}$ ((35.4)–(35.5)); the anticommutators $\sigma^\mu\bar\sigma^\nu + \sigma^\nu\bar\sigma^\mu = -2g^{\mu\nu}$ and $\bar\sigma^\mu\sigma^\nu + \bar\sigma^\nu\sigma^\mu = -2g^{\mu\nu}$ ((36.8)), the diagonal blocks of $\{\gamma^\mu,\gamma^\nu\} = -2g^{\mu\nu}$; and the generators $(S_L^{\mu\nu})_a{}^b = \frac{i}{4}(\sigma^\mu\bar\sigma^\nu - \sigma^\nu\bar\sigma^\mu)_a{}^b$ and $(S_R^{\mu\nu})^{\dot a}{}_{\dot b} = -\frac{i}{4}(\bar\sigma^\mu\sigma^\nu - \bar\sigma^\nu\sigma^\mu)^{\dot a}{}_{\dot b}$ ((35.21)–(35.22)).

Three remarks tie them to the notes. LecNote17 p. 2's first line should read $\bar\sigma^\mu\sigma^\nu + \bar\sigma^\nu\sigma^\mu$. This $(S_R)^{\dot a}{}_{\dot b}$ is LecNote16 p. 10's $(S_R)_{\dot a}{}^{\dot b} = -[(S_L)_a{}^b]^*$ with both indices moved. And S_L is $\frac{i}{2}\sigma_i$ at $\mu\nu = i0$ and $\frac{1}{2}\epsilon^{ijk}\sigma_k$ at ij ((34.9)–(34.10); LecNote16 pp. 8–9); lowering its second index, $(S_L)_{ac} = \varepsilon_{cb}(S_L)_a{}^b$, gives the symmetric form atop LecNote16 p. 14. In four-component form the anticommutators and the generators combine into

$$\gamma^\mu\gamma^\nu = -g^{\mu\nu} - 2iS^{\mu\nu}. \tag{10.26.2}$$

THE DIRAC FIELD, $\bar\Psi$ AND β

$\Psi = (\chi_a, \xi^{\dagger\dot a})$ (LecNote17 p. 4) has a lower undotted and an upper dotted index, so by the table of index positions

$$D(\Lambda) = \begin{pmatrix} L & 0 \\ 0 & (L^\dagger)^{-1} \end{pmatrix} = e^{\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu}}, \qquad S^{\mu\nu} = \begin{pmatrix} S_L^{\mu\nu} & 0 \\ 0 & -(S_R^{\mu\nu})^{\dot a}{}_{\dot b} \end{pmatrix}; \tag{10.26.3}$$

LecNote17 p. 9's $R = L^*$ (on $\psi_a^\dagger$) and §10.25's $R = (L^\dagger)^{-1}$ (on $\xi^{\dagger\dot a}$) are the same representation. The rotation and boost generators are $J_i = \frac{1}{2}\mathrm{diag}(\sigma_i, \sigma_i)$ and $K^i = S^{i0} = \frac{i}{2}\gamma^i\gamma^0 = \frac{i}{2}\mathrm{diag}(\sigma_i, -\sigma_i)$.

Why not $\Psi^\dagger\Psi$**.** $K^\dagger = -K$: boosts are not unitary, and $\Psi^\dagger\Psi \to \Psi^\dagger D^\dagger D\Psi$ is not invariant. It could not be, since $\Psi^\dagger\Psi$ is the charge density, the time component of a vector. β swaps the two blocks, so

$$\beta D^\dagger \beta = D^{-1}, \qquad \text{i.e.} \qquad D^\dagger \beta D = \beta, \tag{10.26.4}$$

and $\bar\Psi \equiv \Psi^\dagger\beta \to \bar\Psi D^{-1}$: $\bar\Psi\Psi$ is a scalar. With the invariance of the symbol γ^μ (LecNote17 p. 2),

$$D^{-1}\gamma^\mu D = \Lambda^\mu{}_\nu \gamma^\nu, \tag{10.26.5}$$

$\bar\Psi\gamma^\mu\Psi$ is a vector and $\mathcal{L} = i\bar\Psi\partial\!\!\!/\Psi - m\bar\Psi\Psi$ a scalar.

β ***versus*** γ^0**.** Same numbers, different objects. β is the invariant (10.26.4) matrix of the pairing $\Psi^\dagger(\cdots)\Psi$; its index structure $\left(\begin{smallmatrix} 0 & \delta_{\dot a}{}^{\dot c} \\ \delta^a{}_c & 0 \end{smallmatrix}\right)$ (LecNote17 p. 5) contracts $\Psi^\dagger = (\chi_{\dot a}^\dagger, \xi^a)$ with Ψ. γ^0 is the $\mu = 0$ component of the vector γ^μ, which by (10.26.5) a boost mixes with the γ^i. As $\beta\gamma^0 = 1$, $\bar\Psi\gamma^0\Psi = \Psi^\dagger\Psi$.

Bars and bilinears. For a 4×4 matrix let $\bar A \equiv \beta A^\dagger \beta$ (LecNote18 p. 6). Then

$$\overline{Au} = \bar u \bar A, \qquad (\bar u_1 A u_2)^* = \bar u_2 \bar A u_1, \qquad (\bar\Psi A\Psi)^\dagger = \bar\Psi \bar A \Psi, \tag{10.26.6}$$

the last for field operators: $\bar A = A$ means a Hermitian bilinear. As $\bar\gamma^\mu = \gamma^\mu$, $\bar S^{\mu\nu} = S^{\mu\nu}$ but $\bar\gamma_5 = -\gamma_5$, the Hermitian bilinears are $\bar\Psi\Psi$, $i\bar\Psi\gamma_5\Psi$, $\bar\Psi\gamma^\mu\Psi$, $\bar\Psi\gamma^\mu\gamma_5\Psi$, $\bar\Psi S^{\mu\nu}\Psi$: two scalars, two vectors and a tensor under D ("pseudo" and "axial" refer to parity). Their 16 matrices span all 4×4 matrices; $\gamma_5 S^{\mu\nu} = \frac{i}{2}\epsilon^{\mu\nu\rho\sigma}S_{\rho\sigma}$ is not new.

γ_5, CHIRALITY AND DUALITY

$\gamma_5 = -\frac{i}{24}\epsilon_{\mu\nu\rho\sigma}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma$ (LecNote17 p. 9) needs $\epsilon_{0123} = -1$; the other sign gives $-\gamma_5$. $\gamma_5^2 = 1$, $\{\gamma_5, \gamma^\mu\} = 0$, and $[\gamma_5, S^{\mu\nu}] = 0$ is what "γ_5 is an invariant symbol" means: $P_L = \frac{1}{2}(1-\gamma_5)$, $P_R = \frac{1}{2}(1+\gamma_5)$ commute with D, so $P_L\Psi = (\chi_a, 0)$ and $P_R\Psi = (0, \xi^{\dagger\dot a})$ transform separately, as $(2,1)$ and $(1,2)$. The eigenvalue of γ_5 is the *chirality*, -1 left- and $+1$ right-handed. Since $\beta\gamma^\mu$ is block diagonal and β is not, $\bar\Psi\partial\!\!\!/\Psi = \chi^\dagger\bar\sigma^\mu\partial_\mu\chi + \xi\sigma^\mu\partial_\mu\xi^\dagger$ keeps the chiralities apart, while $\bar\Psi\Psi = \xi\chi + \chi^\dagger\xi^\dagger$ joins them: only the mass mixes $P_L\Psi$ and $P_R\Psi$.

The same sign fixes LecNote16 p. 17. The dual $(*A)^{\mu\nu} \equiv \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}A_{\rho\sigma}$ of an antisymmetric tensor commutes with proper Lorentz transformations and $** = -1$, so its eigenvalues are $\pm i$, and

$$S_L^{\mu\nu} = -\tfrac{i}{2}\epsilon^{\mu\nu\rho\sigma}S_{L\rho\sigma}, \qquad S_R^{\mu\nu} = +\tfrac{i}{2}\epsilon^{\mu\nu\rho\sigma}S_{R\rho\sigma}. \tag{10.26.7}$$

(p. 17 gives S_R the sign of S_L; for either sign of ϵ^{0123} they are opposite, as $S_R = -S_L^*$.) A real $A^{\mu\nu}$ splits as $G + G^*$, $G = \frac{1}{2}(A - i*A)$ self-dual, the $(3,1)$;

$S_L^{\mu\nu}A_{\mu\nu} = S_L^{\mu\nu}G_{\mu\nu}$ sees only G, the "projection" of p. 17 (for $F^{0i} = E^i$, $F^{ij} = \epsilon^{ijk}B^k$, $G^{0i} = \frac{1}{2}(E - iB)^i$). The same convention gives $\mathrm{Tr}[\gamma_5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = -4i\epsilon^{\mu\nu\rho\sigma}$ (§10.29).

Charge conjugation and Majorana fields

The operator C swaps $\chi \leftrightarrow \xi$ (LecNote17 p. 6); on components it acts through the matrix $C = \mathrm{diag}(\varepsilon_{ac}, \varepsilon^{\dot a\dot c})$ (p. 7):

$$\begin{aligned} C^{-1}\Psi C = \Psi^C \equiv C\bar\Psi^T = (\xi_a, \chi^{\dagger\dot a}), \\ C^T = C^\dagger = C^{-1} = -C, \qquad C^{-1}\gamma^\mu C = -(\gamma^\mu)^T. \end{aligned} \tag{10.26.8}$$

The last identity (p. 8) does two jobs. First, through $C^{-1}S^{\mu\nu}C = -(S^{\mu\nu})^T$ it gives $C(D^{-1})^TC^{-1} = D$, so Ψ^C transforms like Ψ. Second, it makes Ψ^C a solution whenever Ψ is one: transpose the conjugate equation, $i\partial_\mu\bar\Psi\gamma^\mu + m\bar\Psi = 0$, and multiply by C. For plane waves, with LecNote18 p. 5's phases, exactly

$$v_s(\mathbf{p}) = C\bar u_s(\mathbf{p})^T, \qquad u_s(\mathbf{p}) = C\bar v_s(\mathbf{p})^T, \tag{10.26.9}$$

at rest and after the boost; this fixes the phases of the v's, which the Dirac equation leaves free. C-invariance is not charge conservation: $\chi \leftrightarrow \xi$ sends $j^\mu = \bar\Psi\gamma^\mu\Psi = \chi^\dagger\bar\sigma^\mu\chi - \xi^\dagger\bar\sigma^\mu\xi$ (LecNote17 p. 6; the minus is (10.26.1)) to $-j^\mu$. Charge conservation is the $U(1)$ $\Psi \to e^{-i\alpha}\Psi$.

Majorana. For one Weyl field with mass term $-\frac{1}{2}m(\psi\psi + \psi^\dagger\psi^\dagger)$ (LecNote17 p. 1), $\psi \to e^{-i\alpha}\psi$ is no symmetry: it multiplies $\psi\psi$ by $e^{-2i\alpha}$. The four-component form $\Psi = (\psi_a, \psi^{\dagger\dot a})$ (p. 3) is $\chi = \xi = \psi$, the Majorana condition $\Psi^C = \Psi$. Then $\bar\Psi = \Psi^TC$, and $\mathcal{L} = \frac{i}{2}\bar\Psi\partial\!\!\!/\Psi - \frac{m}{2}\bar\Psi\Psi = \frac{i}{2}\Psi^TC\gamma^\mu\partial_\mu\Psi - \frac{m}{2}\Psi^TC\Psi$ (p. 8). The $\frac{1}{2}$'s undo a doubling: $\bar\Psi\Psi = \psi\psi + \psi^\dagger\psi^\dagger$, and by (10.26.1) $\bar\Psi\partial\!\!\!/\Psi = 2\psi^\dagger\bar\sigma^\mu\partial_\mu\psi$ plus a total derivative. These terms survive because C is antisymmetric and $C\gamma^\mu$ symmetric: for anticommuting Ψ, $\Psi^TM\Psi = -\Psi^TM^T\Psi$ kills a symmetric M, and $\Psi^TB\partial\Psi = \Psi^TB^T\partial\Psi - \partial(\Psi^TB^T\Psi)$ makes an antisymmetric B a total derivative.

Plane waves at rest and boosted

$\Psi = u\,e^{ipx} + v\,e^{-ipx}$, $p^0 = \omega > 0$, needs $(p\!\!\!/ + m)u = 0$ and $(-p\!\!\!/ + m)v = 0$ (LecNote18 p. 4). On shell $p\!\!\!/^2 = m^2$, so $\Pi_\pm = (m \mp p\!\!\!/)/2m$ are complementary projectors of trace 2, $(p\!\!\!/ + m)\Pi_+ = 0 = (-p\!\!\!/ + m)\Pi_-$: two u's and two v's (Klein–Gordon is necessary, not sufficient). At rest $p_\mu = (-m, \mathbf{0})$, $p\!\!\!/ = -m\gamma^0$, and

$$p\!\!\!/ + m = m\begin{pmatrix} I & -I \\ -I & I \end{pmatrix}, \qquad -p\!\!\!/ + m = m\begin{pmatrix} I & I \\ I & I \end{pmatrix} \tag{10.26.10}$$

(p. 4 writes the first with the opposite sign; same null space).
$S_z = S^{12} = \frac{1}{2}\mathrm{diag}(\sigma_3, \sigma_3)$ picks the basis of p. 5,

$$u_s(\mathbf{0}) = \sqrt{m}\begin{pmatrix}\xi_s\\ \xi_s\end{pmatrix}, \qquad v_s(\mathbf{0}) = \sqrt{m}\begin{pmatrix}\zeta_s\\ -\zeta_s\end{pmatrix}, \tag{10.26.11}$$

where $\xi_+ = (1,0)^T$, $\xi_- = (0,1)^T$, $\zeta_+ = (0,1)^T$, $\zeta_- = (-1,0)^T$, with $S_z u_\pm = \pm\frac{1}{2}u_\pm$ but $S_z v_\pm = \mp\frac{1}{2}v_\pm$. Why $\mp$: the antiparticle operator $d_s^\dagger$ is projected out of Ψ with $v_s^\dagger$ (LecNote18 pp. 9, 13), and $[J_z, \Psi] = -S_z\Psi$ up to an orbital term that integrates away at $\mathbf{p} = \mathbf{0}$; so the antiparticle's spin is minus the S_z of v_s, and v_+ describes spin up.

The boost. With $\sinh\eta = |\mathbf{p}|/m$, $u_s(\mathbf{p}) = e^{i\eta\hat{\mathbf{p}}\cdot\mathbf{K}}u_s(\mathbf{0})$ and $v_s(\mathbf{p}) = e^{i\eta\hat{\mathbf{p}}\cdot\mathbf{K}}v_s(\mathbf{0})$ (LecNote18 p. 6) solve the equations at $\mathbf{p}$, since (10.26.5) gives $D\not{p}(\mathbf{0})D^{-1} = \not{p}$. In the Weyl basis $e^{i\eta\hat{\mathbf{p}}\cdot\mathbf{K}} = \mathrm{diag}(e^{-\eta\hat{\mathbf{p}}\cdot\boldsymbol{\sigma}/2}, e^{\eta\hat{\mathbf{p}}\cdot\boldsymbol{\sigma}/2})$ and $\sqrt{m}\,e^{\mp\eta\hat{\mathbf{p}}\cdot\boldsymbol{\sigma}/2} = \sqrt{\omega \mp \mathbf{p}\cdot\boldsymbol{\sigma}}$, so

$$u_s(\mathbf{p}) = \begin{pmatrix}\sqrt{-p\cdot\sigma}\,\xi_s\\ \sqrt{-p\cdot\bar\sigma}\,\xi_s\end{pmatrix}, \qquad v_s(\mathbf{p}) = \begin{pmatrix}\sqrt{-p\cdot\sigma}\,\zeta_s\\ -\sqrt{-p\cdot\bar\sigma}\,\zeta_s\end{pmatrix}, \tag{10.26.12}$$

with $-p\cdot\sigma = \omega - \mathbf{p}\cdot\boldsymbol{\sigma}$, $-p\cdot\bar\sigma = \omega + \mathbf{p}\cdot\boldsymbol{\sigma}$ (positive matrices, positive roots); with §10.25's K the same exponential would give $u(-\mathbf{p})$. K is anti-Hermitian and $\bar K = K$, so $\bar u_s(\mathbf{p}) = \bar u_s(\mathbf{0})e^{-i\eta\hat{\mathbf{p}}\cdot\mathbf{K}}$; barring the equations gives $\bar u(\not{p} + m) = 0$, $\bar v(-\not{p} + m) = 0$.

Normalizations, orthogonality and the Gordon identity

$\bar u_{s'}u_s$ is invariant and $\bar u_{s'}\gamma^\mu u_s$ a vector, so the rest frame gives, at equal momenta (LecNote18 pp. 7, 9),

$$\begin{gathered}\bar u_{s'}u_s = 2m\delta_{s's}, \qquad \bar v_{s'}v_s = -2m\delta_{s's}, \qquad \bar u_{s'}v_s = \bar v_{s'}u_s = 0,\\ \bar u_{s'}\gamma^\mu u_s = \bar v_{s'}\gamma^\mu v_s = 2p^\mu\delta_{s's}.\end{gathered} \tag{10.26.13}$$

At $\mu = 0$, $u^\dagger u = v^\dagger v = 2\omega$; that is why $u(\mathbf{0})$ carries $\sqrt{m}$: it gives $\{b, b^\dagger\}$ the scalar states' $(2\pi)^3 2\omega\delta^3$ (LecNote18 p. 12, §10.27), and $\bar u u = 2m$ is its invariant form.

Opposite momenta. Inverting the mode expansion (LecNote18 p. 8) needs, as boxed on p. 9,

$$\bar u_{s'}(\mathbf{p})\gamma^0 v_s(-\mathbf{p}) = 0 = \bar v_{s'}(\mathbf{p})\gamma^0 u_s(-\mathbf{p}). \tag{10.26.14}$$

Proof: $\bar u\gamma^0 = u^\dagger$; as $K^\dagger = -K$, $u_{s'}(\mathbf{p})^\dagger = u_{s'}(\mathbf{0})^\dagger e^{i\eta\hat{\mathbf{p}}\cdot\mathbf{K}}$, while $v_s(-\mathbf{p}) = e^{-i\eta\hat{\mathbf{p}}\cdot\mathbf{K}}v_s(\mathbf{0})$; the product is $u_{s'}(\mathbf{0})^\dagger v_s(\mathbf{0}) = 0$. The γ^0 matters: $\bar u_{s'}(\mathbf{p})v_s(-\mathbf{p}) \neq 0$. The same steps give $\bar u_{s'}(\mathbf{p})\gamma^0 u_s(-\mathbf{p}) = 2m\delta_{s's}$, not p. 12's 0; p. 12's relations should be (10.26.14).

The Gordon identity. By (10.26.2), $\gamma^\mu \not{p} = -p^\mu - 2iS^{\mu\nu}p_\nu$ and $\not{p}'\gamma^\mu = -p'^\mu + 2iS^{\mu\nu}p'_\nu$ (LecNote18 p. 11 drops the last prime). With $\not{p}u = -mu$ and $\bar{u}'\not{p}' = -m\bar{u}'$,

$$2m\,\bar{u}_{s'}(\mathbf{p}')\gamma^\mu u_s(\mathbf{p}) = \bar{u}_{s'}(\mathbf{p}')\big[(p+p')^\mu - 2iS^{\mu\nu}(p'-p)_\nu\big]u_s(\mathbf{p})\,: \tag{10.26.15}$$

a convection term, the $(p+p')^\mu$ of a Klein–Gordon current between plane waves, plus a spin term. At $p'=p$ it turns $\bar{u}u = 2m$ into $\bar{u}\gamma^\mu u = 2p^\mu$. For v the right side changes sign.

Spin sums

$$\sum_{s=\pm} u_s(\mathbf{p})\bar{u}_s(\mathbf{p}) = -\not{p} + m, \qquad \sum_{s=\pm} v_s(\mathbf{p})\bar{v}_s(\mathbf{p}) = -\not{p} - m \tag{10.26.16}$$

(LecNote18 p. 7) follow in two ways. (a) At rest $\sum_s u_s\bar{u}_s = m(1+\gamma^0) = -\not{p}(\mathbf{0}) + m$ (p. 7's $-m\gamma^0 + m$ is a sign slip), and conjugating with the boost sends γ^0 to $-\not{p}/m$. (b) Without components: $X = \sum_s u_s\bar{u}_s$ has $Xu_t = 2mu_t$ and $Xv_t = 0$ by (10.26.13), and so has $-\not{p} + m$; the four u_t, v_t are independent, so the two agree. Subtracting, $\sum_s(u_s\bar{u}_s - v_s\bar{v}_s) = 2m$. The u sum is the numerator of $\bar{S}(p) = (-\not{p} + m)/(p^2 + m^2 - i\epsilon)$, and the sums turn a spin-summed $|\mathcal{T}|^2$ into a trace (§10.29).

Helicity and chirality

Helicity is the spin along the momentum, $\hat{\mathbf{p}}\cdot\mathbf{S}$ with $S^i = \frac{1}{2}\epsilon^{ijk}S^{jk} = \frac{1}{2}\mathrm{diag}(\sigma_i, \sigma_i)$. Chirality is Lorentz invariant but the mass mixes it; the helicity of a massive particle depends on the frame ($[K^3, S_z] = 0$: a boost along z that overtakes the particle reverses $\mathbf{p}$ and keeps S_z). At $m = 0$ the two coincide. With $u = (u_L, u_R)$, $\not{p}u = 0$ splits into $(p\cdot\sigma)u_R = 0$ and $(p\cdot\bar{\sigma})u_L = 0$, i.e. $\hat{\mathbf{p}}\cdot\boldsymbol{\sigma}\,u_{R,L} = \pm u_{R,L}$: on massless solutions $2\hat{\mathbf{p}}\cdot\mathbf{S} = \gamma_5$. A right-handed u has helicity $+\frac{1}{2}$; a v describes an antiparticle with minus the spinor's spin, so a positron of helicity $+\frac{1}{2}$ has a left-handed v. (Here, as above, right- and left-handed name the chirality, as for Srednicki's Weyl fields in §34; in §38 he also uses the words for helicity, calling helicity $+\frac{1}{2}$ right-handed.) For $m \neq 0$ and $\mathbf{p} = p\hat{\mathbf{z}}$, $p > 0$, (10.26.12) gives $u_+ = (\sqrt{\omega - p}\,\xi_+, \sqrt{\omega + p}\,\xi_+)$ and $u_- = (\sqrt{\omega + p}\,\xi_-, \sqrt{\omega - p}\,\xi_-)$: the wrong chirality has relative weight $\sqrt{(\omega - p)/2\omega} \simeq m/2\omega$. In the projector form of LecNote18 p. 8 (skipped in class), exactly

$$u_s\bar{u}_s = \tfrac{1}{2}(1 - s\gamma_5\not{z})(-\not{p} + m), \qquad v_s\bar{v}_s = \tfrac{1}{2}(1 - s\gamma_5\not{z})(-\not{p} - m),$$

with $z^\mu = (p/m, 0, 0, \omega/m)$ the boosted $\hat{\mathbf{z}}$. As $m \to 0$ at fixed p, $z^\mu = p^\mu/m + O(m/p)$, and the two tend to $\frac{1}{2}(1 + s\gamma_5)(-\not{p})$ and $\frac{1}{2}(1 - s\gamma_5)(-\not{p})$ with corrections of order m.